

Solution Requirements

FitBuddy – AI Fitness Plan Generator using Gemini Models

Functional Requirements

- Accept name, user ID, age, weight, fitness goal and workout intensity.
- Generate personalized 7-day workout plan.
- Generate nutrition/recovery tip.
- Store user and plan information in SQLite via SQLAlchemy.
- Accept feedback and update the workout plan.
- Display original/updated plans in an administrative view.
- Provide FastAPI routes and an interactive HTML interface.

Non-Functional Requirements

- Clear and usable web interface.
- Responsive frontend layout.
- Modular backend/frontend organization.
- Environment-based Gemini API key configuration.
- Maintainable code layout.
- Local execution through FastAPI/Uvicorn.

Project Users

Fitness-plan user and administrator/coach view.